

Scientific Notation - The Lock

Solve each problem and use the solutions to unlock the code by substituting the answers into the unlocking code.

1) The space shuttle program costs \$5800 million. Write this number in scientific notation. Use the value of the power for a in the unlocking code.	 $a =$
2) An old unit of measurement called a fluid ounce is equal to 0.0296 litres. Write this in scientific notation. Use the value of the power for b in the unlocking code.	 $b =$
3) The largest ocean in the world is the Pacific Ocean. Its area is approximately equal to $1.813 \times 10^8 \text{ km}^2$. Write this number out in full. Divide this number by 7×10^5 to obtain the value for c in the unlocking code.	 $c =$
4) A particle is radioactive for 2.3×10^{-4} seconds. Write this number out in full. The number of zeros in the number is the value of d in the unlocking code.	 $d =$
5) The diameter of a red blood cell is $6.5 \times 10^{-3} \text{ mm}$. Write this number out in full. Multiply this by 10000 to obtain the value of e in the unlocking code.	 $e =$

6) A group of 8 workers won a syndicate worth £36 million which was shared equally amongst them. How much does each person receive? Write this number in scientific notation. The value of the power is the value you should use for f in the unlocking code.



f =

7) A compact disc can store 1.44×10^6 bytes of information. How many bytes of information can 25 of these discs store? Express your answer in standard form. Use the value of the power for g in the unlocking code.



g =

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$$\frac{e^2 + cd - ab + g}{f}$$

Answers - Scientific Notation: The Lock

$$a = 9$$

$$b = -2$$

$$c = 259$$

$$d = 4$$

$$e = 65$$

$$f = 6$$

$$g = 7$$

Lock: 881